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Refracturing and Horizontal Well Reconstruction to Enhance Recovery in Legacy Shale Oil Wells: A Case Study from the Xi233 Block, Ordos Basin

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Abstract

This study addresses declining productivity, casing integrity issues, and reservoir uncertainty in legacy shale oil wells in the Ordos Basin, with a focus on the Xi233 block. To enhance recovery and stabilize production, the feasibility of refracturing and horizontal well reconstruction was evaluated using an integrated framework combining wellbore diagnostics, distributed fiber-optic sensing, and geological-engineering modeling. Mechanical diagnostics identified well integrity issues, which were resolved using casing-in-casing cementing, while coiled tubing–deployed fiber-optic sensors and pressure gauges characterized pressure support and fluid intake. A coupled geologic and engineering model guided stage-level refracturing design, optimizing fluid volumes and fracture spacing. Field implementation confirmed that the approach restored wellbore integrity and improved treatment efficiency, leading to symmetrical bi-wing fracture propagation and significantly expanded stimulated reservoir volume (SRV). Post-treatment production in the Xi233 block increased from 1.5 t/d to over 10 t/d, demonstrating the technical and economic viability of the workflow. As one of the first integrated case studies in the Ordos Basin, this work offers a repeatable, scalable solution for revitalizing aging shale wells and provides valuable insights for redevelopment strategies in similar basins worldwide.

Introduction

The Ordos Basin, China's second-largest oil- and gas-bearing basin, hosts the Qingcheng oilfield, the country's largest proven shale oil field with reserves exceeding 1 billion tons. Since development began in 2004, the field has gone through early exploration, pilot testing, and scaled-up production. To date, over 1,400 horizontal shale oil wells have been drilled, forming a production base exceeding 3 million tons annually. Technological evolution in shale oil stimulation has progressed from early-stage hydraulic jetting and annular proppant injection to bridge plug and perforation-based cluster fracturing, and more recently to fine-scale segmentation using dissolvable ball seat technology. However, early shale wells were constrained by limited equipment and conceptual understanding, resulting in low fracture complexity and poor reservoir

contact due to pronounced vertical and lateral heterogeneity. Recovery factors remained between 3% and 4%, leaving significant residual reserves as indicated by dynamic monitoring.

Refracturing has emerged as a key technology for revitalizing such wells by recharacterizing the reservoir, redesigning the stimulation treatment, and establishing a complex fracture network to enhance EUR. Facing common challenges such as poor fracture complexity, proppant embedment, and casing deformation, major refracturing techniques include diverting agents, expandable liner patches, and casing-in-casing reconstruction. Of these, diverting-based refracturing has seen broad adoption in Daqing and Changqing, while expandable liners have proven effective in Mahu tight conglomerates. The casing-in-casing refracturing method has been used mainly in shale gas wells, with few shale oil applications reported in China. This study focuses on applying casing-in-casing reconstruction technology to legacy shale oil horizontal wells in the Xi233 block of Changqing Oilfield, aiming to verify its technical feasibility and production benefits through field trials.

Geological and Engineering Challenges in the Xi233 Shale Oil Block

Geological Background. Located in the southwestern Ordos Basin, the Xi233 block targets the Chang 71 and Chang 72 members of the Triassic Yanchang Formation. The reservoir is characterized by semi-deep to deep lacustrine facies, with lithological assemblages dominated by mudstone interbedded with thin layers of fine sandstone (classified as Type II). The source rocks are primarily dark shale and mudstone of the Chang 73 sub-member, with thicknesses ranging from 10 to 40 m and total organic carbon (TOC) contents between 3% and 28%. The effective reservoir thickness ranges from 6 to 12 m, with individual layers of 3 to 5 m. Porosity is between 6% and 12%, and permeability is extremely low, typically between 0.01 and 0.10 mD. The formation pressure coefficient ranges from 0.7 to 0.8, and the crude oil viscosity is 1–3 mPa·s. Overall, the formation exhibits typical shale oil reservoir characteristics: ultra-low permeability, thin reservoir layers, and low oil viscosity.

Horizontal Well Development Status.

Low Productivity and Efficiency

Across the basin, more than 1,400 shale oil horizontal wells yield an average of 5.3 t/d per well. In the Xi233 block, legacy wells drilled between 2011 and 2013 produce only 1.7 t/d per well on average. Taking the representative XP3 well as an example, it was completed in 2012, targeting the Chang 721 reservoir with a 1,535 m horizontal section and a 97% reservoir encounter rate. The initial completion included 11 stages. At startup in December 2012, the well produced 17.54 m³/d of fluid and 12.72 t/d of oil, with 14.7% water cut. Before refracturing, production declined to 6.18 m³/d of fluid and 1.5 t/d of oil, with the water cut rising to 75.7%, indicating low-efficiency production.

Insufficient Initial Stimulation

XP3's initial stimulation involved 11 stages, injecting 6,559.5 m³ of fluid and 483.4 m³ of proppant, with an average sand ratio of 10.5% and a pumping rate of 6 m³/min. In comparison, new shale oil wells in eastern Gansu use over 1,300 m³ of proppant in more than 20 stages. Thus, the initial stimulation scale in XP3 was relatively small. Simulations and monitoring indicate a 4.7% drop in oil saturation, with residual oil concentrated between fractures and in the toe section, highlighting substantial untapped potential and the urgent need for restimulation.

Wellbore Integrity Failure

Two types of wellbore integrity failure affect shale oil wells: structural and sealing failures. Structural failure leads to casing deformation, hindering bridge plug setting and increasing refracturing risks. Sealing failures manifest as sustained casing pressure in the annulus, posing significant well control hazards.

Low Formation Energy

Prolonged depletion in horizontal wells has led to substantial formation energy loss. Low-pressure zones near old fractures may redirect new fractures during refracturing, causing interference between stages or

wells. A common mitigation method, both domestically and internationally, is high-volume fluid preloading before refracturing. This approach enhances formation energy, stabilizes production, and helps restore the stress differential and fracture propagation environment to promote complex fracture networks.

Refracturing Techniques

Three main refracturing techniques are employed: diverting, mechanical isolation, and coiled tubing–assisted methods. For legacy wells like XP3, diverting agents can cause uncontrolled fluid distribution and reservoir damage. Coiled tubing methods are costly, while mechanical isolation requires high wellbore integrity. By contrast, the casing-in-casing technique has demonstrated superior adaptability in field applications. This method not only restores the mechanical integrity of the old wellbore but also provides reliable zonal isolation across extended damaged intervals, enabling high-rate injection during volumetric refracturing. Field practices in the Changqing Oilfield have confirmed that casing-in-casing reconstruction is a cost-effective and technically robust solution, especially for horizontal shale oil wells with narrow annuli and complex deformation conditions (Casing-in-casing wellbore reconstruction and volumetric refracturing technology of horizontal well in Changqing Oilfield, 2023).

Pre-Fracturing Integrated Diagnostics. A comprehensive diagnostic program was conducted for Well XP3 using a combination of mechanical logging, microseismic monitoring, pressure buildup testing, and fiber-optic distributed injection profiling. The objective was to assess wellbore conditions, initial fracture geometry, formation energy, and segment-level fluid intake behavior along the lateral section.

Mechanical Logging

Mechanical logging was conducted using an MIT-MTT combination tool deployed via coiled tubing. The results identified the perforation status, casing corrosion, and structural anomalies. As shown in Fig. 1, red indicates enlarged inner diameters, blue indicates restrictions, and green represents normal casing diameter. No significant casing deformation was observed, indicating intact structural integrity. However, localized severe corrosion (Fig. 1a), casing perforation (Fig. 1b), and scale buildup (Fig. 1c) were identified, indicating compromised sealing integrity and the need for wellbore reconstruction.

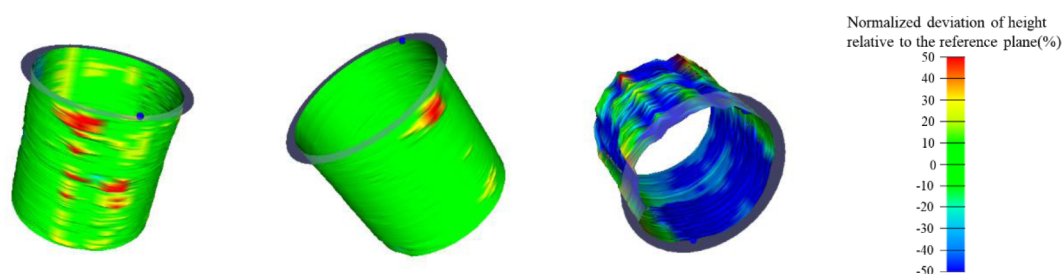


Figure 1—Typical imaging results from MIT-MTT logging of XP3 showing wellbore integrity failures:(a) Severe corrosion; (b) Casing perforation; (c) Heavy scaling.

Microseismic Monitoring

Microseismic monitoring of six stages from the initial completion revealed that fracture events were distributed unevenly along the lateral, as shown in Fig. 2. The fractures exhibited banded distributions with clear gaps between events, indicating limited stimulated volume and inadequate fracture complexity. These observations justify refracturing to reconstruct a more effective fracture network.

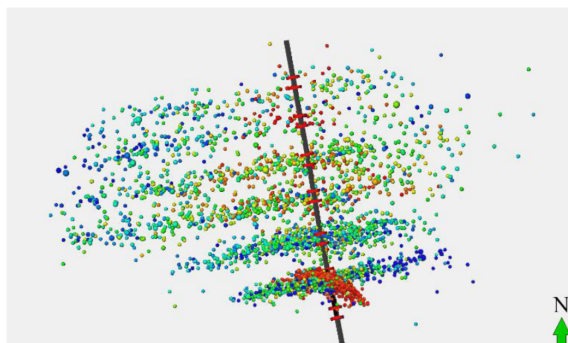


Figure 2—Microseismic mapping results from the initial stimulation of XP3 showing sparse and unevenly distributed fracture events.

Pressure Buildup Testing

The original formation pressure in the Chang 7 reservoir was 15.8 MPa. A pressure buildup test conducted in 2024 showed a 2.48 MPa recovery over 380 hours, with a final pressure of 9.99 MPa. Pressure matching yielded an estimated formation pressure of 13.48 MPa, corresponding to 85% of the original pressure level. These results confirm partial depletion and the necessity for energy restoration prior to refracturing (see Fig. 3).

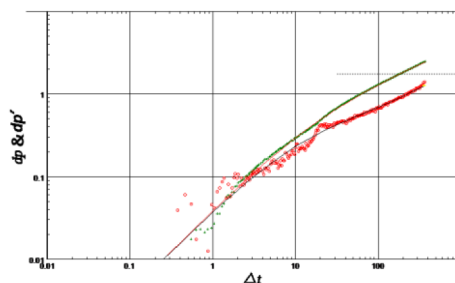


Figure 3—Pressure buildup test curve of XP3 showing partial formation energy depletion and indicating the need for energy supplementation.

Fiber-Optic Distributed Injection Profiling

A distributed temperature sensing (DTS) test was conducted via coiled tubing, injecting 540 m³ of water over 4 hours and measuring temperature recovery post-shutdown to infer segmental fluid intake. The results revealed significant heterogeneity: Stages 8–11 had the highest intake, Stages 3–7 had moderate intake, while Stages 1–2 showed minimal fluid entry (see Table 1). Higher intake correlates with lower formation energy and higher depletion. Intake was concentrated near the heel, while the toe was relatively inactive, indicating strong lateral heterogeneity in reservoir permeability and the need to reconstruct the flow field via refracturing.

Table 1—Segmental fluid intake percentages for different fracturing stages in Well XP3.

Stage	Intake (%)	Volume (m ³)	Stage	Intake (%)	Volume (m ³)
1	0.11%	0.59	7	7.25%	39.05
2	0.18%	0.99	8	13.81%	74.46
3	1.85%	9.97	9	21.07%	113.55
4	2.15%	11.58	10	18.31%	98.67
5	4.22%	22.72	11	26.25%	141.48
6	4.81%	25.93	—	—	—

Integrated Diagnostic Summary

Integrated diagnostics revealed several challenges: sealing integrity failure, limited initial fracture coverage, low formation energy, and severe intake imbalance along the lateral. These findings support the implementation of casing reconstruction and refracturing to restore wellbore conditions, reconstruct the fracture network, and rebuild the pressure, stress, and flow fields, ultimately aiming to rejuvenate production in this legacy horizontal well.

Integrated Refracturing Technologies for Horizontal Wells

Casing Reconstruction for Refracturing. Pre-fracturing diagnostics indicated significant wellbore degradation in horizontal shale wells due to long-term production, including severe corrosion and perforation. Traditional refracturing methods pose high operational risks in such wells. The casing-in-casing reconstruction technique offers an effective solution for restoring wellbore integrity in legacy wells. This method involves running a smaller-diameter casing (OD 114.3 mm, ID 94 mm) inside the original casing (OD 139.7 mm, ID 124.26 mm) followed by cementing, thereby enabling modern plug-and-perf staged fracturing with fine segmentation.

Compared to conventional dual-packer methods—which suffer from high failure rates, low efficiency, and limitations due to casing damage or original completion structure—this approach allows precise zonal refracturing, maximizing remaining hydrocarbon recovery. It also offers high completion flexibility, better safety, and lower overall cost. The technique has already shown promising results in both Changqing and Southwest Oilfields.

The feasibility of small casing deployment was assessed using the Landmark engineering suite. A mechanical model was developed to simulate the running process down to 3,745 m measured depth. Dynamic variations in hook load, effective stress, and frictional resistance were analyzed with depth. The results confirmed that XP3 satisfies all mechanical conditions required for full-length casing reconstruction.

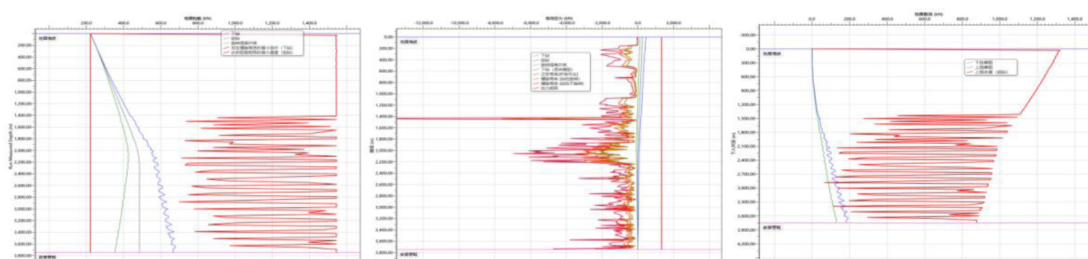


Figure 4—Feasibility analysis for small-diameter casing deployment in Well XP3:
(a) Hook load analysis; (b) Effective stress profile; (c) Frictional drag evaluation.

Optimization of Energy Supplementation Segments and Volume. To rebuild the pressure and stress fields prior to refracturing, a fiber-optic distributed profiling test was conducted. As shown in Fig. 5, fluid intake was concentrated near the heel, while the toe showed low intake, indicating higher depletion near the heel and a greater need for energy replenishment.

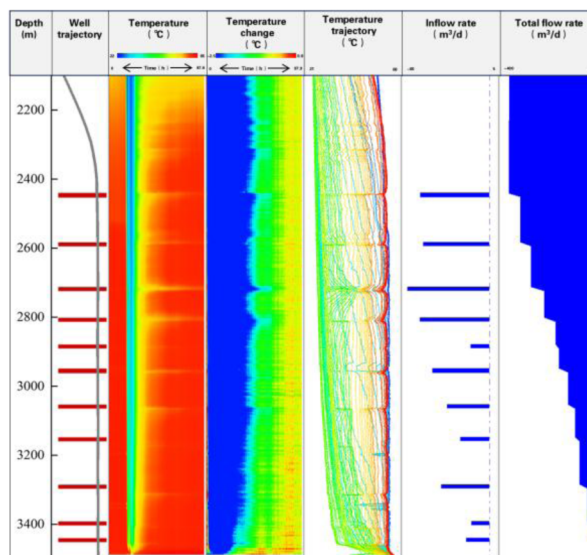


Figure 5—Fiber-optic distributed injection profiling of XP3 showing fluid intake concentration at the heel and lower activity at the toe.

Based on these findings, a segment-specific energy supplementation strategy was developed, as shown in Fig. 6. The plan includes four large injection segments, with a total volume of 8,600 m³. The goal was to uniformly increase formation pressure along the lateral back to the original 15.8 MPa. This also raised net pressure within existing fractures, reducing the risk of fracture interference during subsequent refracturing.

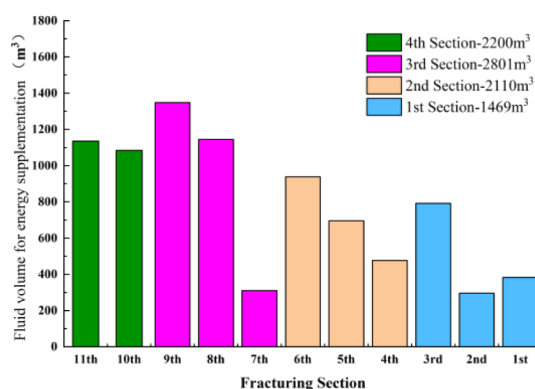


Figure 6—Optimized energy supplementation plan for XP3 showing staged fluid injection volumes and pressure-balancing objectives along the lateral.

Optimization of Perforation Clusters for Refracturing. Geological and engineering "sweet spots" were identified by integrating multi-source logging data and applying a comprehensive sweet spot calculation algorithm. This was overlaid with initial perforation locations to generate a composite sweet spot map for refracturing (Fig. 7). Based on the 3D distribution of residual oil and pulse neutron logging results, the refracturing strategy focused on denser clustering in Class II toe sections (with slightly lower quality) and supplemental targeting of Class I heel sections. A total of 18 segments were selected, with 2–3 clusters per segment for a total of 43 perforation clusters. A dynamic, multi-stage diverting method was designed to ensure effective stimulation across all clusters.

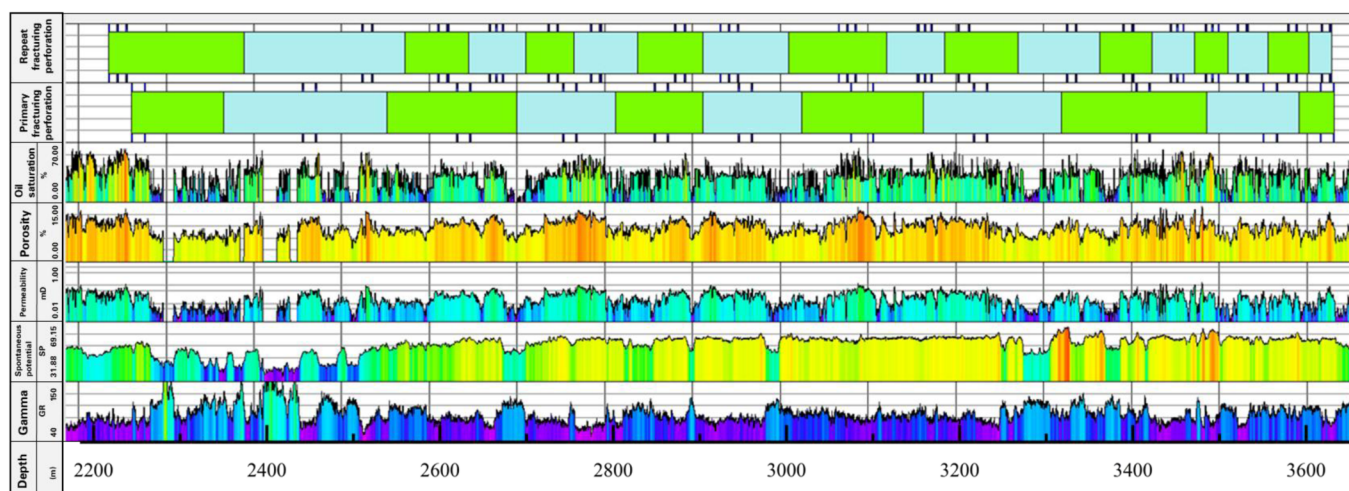
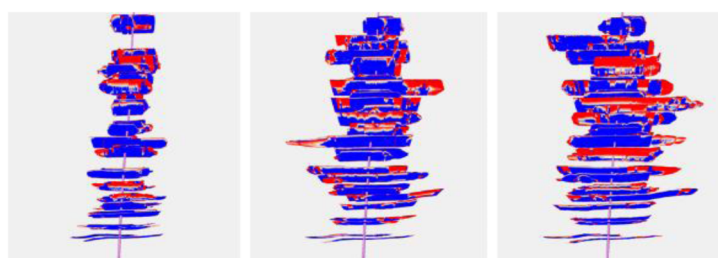


Figure 7—Integrated sweet spot map for refracturing in XP3 showing optimized perforation clusters based on residual oil and geological attributes.

Optimization of Refracturing Parameters. With the perforation cluster locations defined, a full-lifecycle geo-engineering integrated model was applied to perform a sensitivity analysis on key refracturing parameters. Simulations were conducted by varying the fluid volume per stage (500–1,200 m³) and proppant volume per stage (80–160 m³), keeping other parameters constant.

As shown in Fig. 8, increased fluid and proppant volumes per stage led to greater fracture complexity and stimulated reservoir volume (SRV). Specifically:

- Fig. 8a–c illustrate how increasing fluid volume from 500 m³ to 1,200 m³ per stage enhances fracture length and branch development.
- Fig. 8d–f show the effect of increasing proppant volume from 80 m³ to 140 m³ per stage, resulting in wider and more conductive fracture networks.



Further analysis of SRV response under different input intensities is presented in Fig. 9:

- When fluid volume exceeds 1,000 m³ per stage, the SRV growth rate plateaus, indicating diminishing returns.
- When proppant exceeds 120 m³ per stage, further increases contribute little to conductivity improvement.

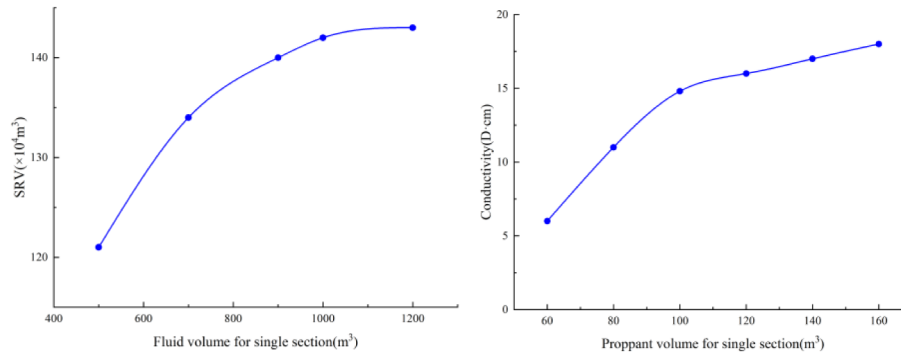


Figure 9—Impact of fluid volume (left) and proppant volume (right) per stage on stimulated reservoir volume (SRV), showing optimal thresholds for efficient fracture stimulation.

Thus, the optimized design recommends approximately 1,000 m³ of fluid and 120 m³ of proppant per stage for balanced efficiency and cost-effectiveness.

Field Implementation of Horizontal Well Refracturing

Field Execution. Based on the optimized refracturing design, a high-efficiency volumetric refracturing technique was implemented to enhance stimulation outcomes:

1. To address strong reservoir heterogeneity and scaling tendencies, an integrated fracturing–displacement–anti-scaling fluid system was used.
2. Extreme rate-limited fracturing and multi-stage composite diverting techniques were applied to ensure effective fracture propagation under safe surface pressure limits.
3. Global refracturing practices indicate that micro-proppants (>100 mesh) can enter microfractures in shale formations, increasing effective fracture volume. Therefore, a combination of 70/140 mesh and 20/40 mesh proppants was selected to optimize in-fracture proppant distribution.
4. Real-time microseismic monitoring was arranged in a nearby observation well to facilitate dynamic treatment adjustments.

The target well, XP3, underwent successful plug-and-perf refracturing in 18 stages with 43 clusters. As shown in Fig. 10, the treatment maintained stable pressures throughout, with clear pressure responses after temporary plugging events. The average proppant intensity was 8 t/m, and fluid intensity was 67.8 m³/m.

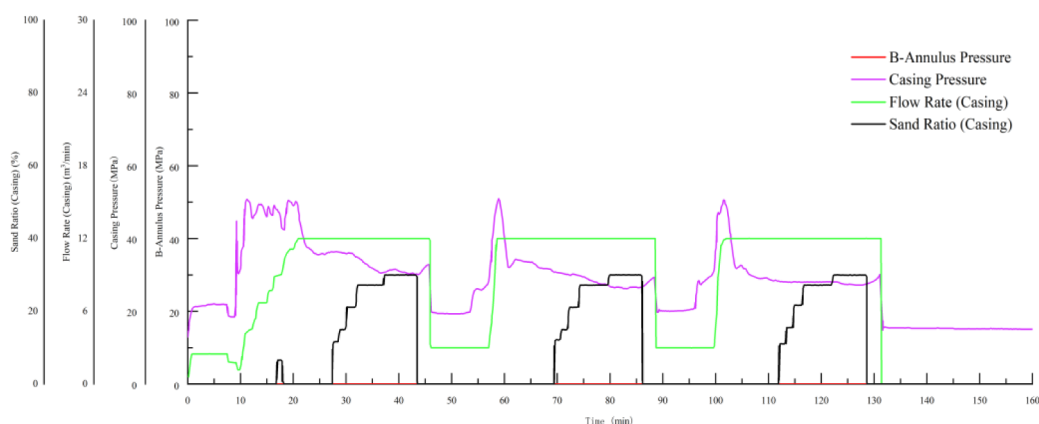


Figure 10—Fracture pressure curve for Stage 11 of XP3 during refracturing, showing stable treatment pressure and clear response to in-fracture temporary plugging.

Microseismic monitoring results are shown in Fig. 11. The fractures had half-lengths of 150–200 m, heights of 40–50 m, and widths of 50–80 m. The dominant geometry was symmetrical bi-wing fractures, forming a large stimulated reservoir volume (SRV) with effective fracture spacing control across all stages.

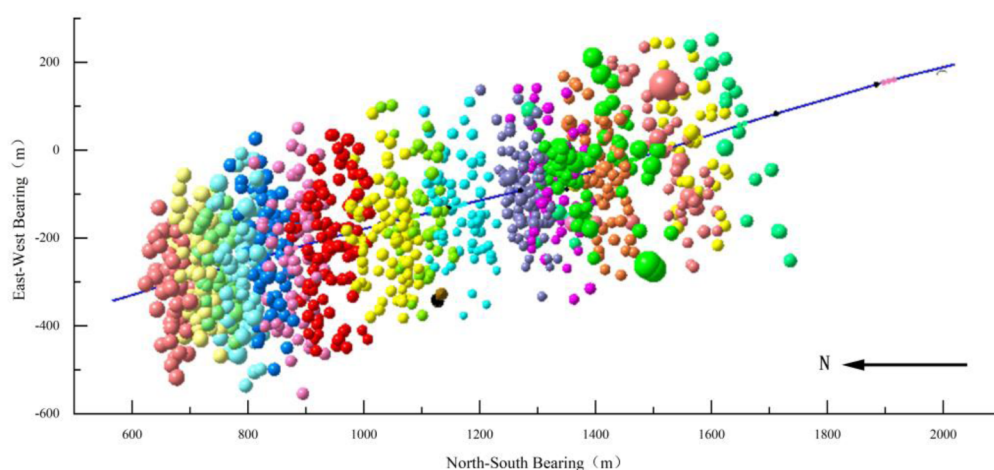


Figure 11—Microseismic imaging results of XP3 refracturing showing bi-wing fracture geometry and extensive SRV development.

Simulation-Based Refracturing Effectiveness Evaluation. Based on the microseismic data, an embedded discrete fracture modeling (EDFM) approach was used to calibrate the numerical fracture propagation model. A stage-by-stage intelligent fitting of treatment pressure was performed using the microseismic events as constraints.

As shown in Fig. 12, the simulation achieved over 90% match accuracy with actual pump-in pressure, validating the calibration quality.

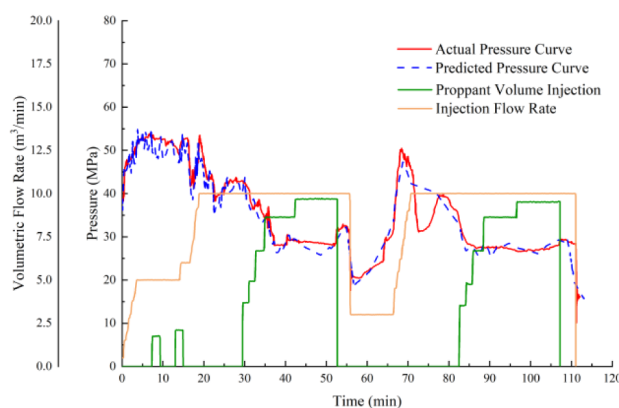


Figure 12—Intelligent fitting of treatment pressure using EDFM constrained by microseismic data, showing >90% match accuracy across refracturing stages.

The simulated fracture geometry is shown in Fig. 13, and it aligns closely with observed microseismic event distributions. The model yielded average hydraulic fracture half-lengths of 158.6 m and propped half-lengths of 138.8 m, with an SRV-to-DRV ratio of 87.5% and average fracture conductivity of 16.8 D·cm. These results confirm that the refracturing treatment met the design targets.

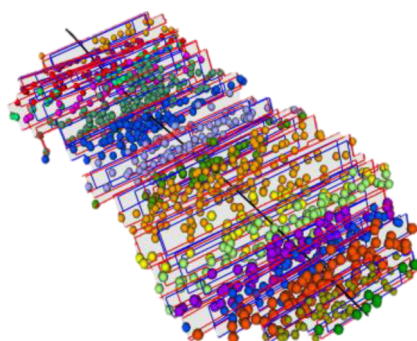


Figure 13—Simulated fracture geometry and extent of XP3 refracturing based on calibrated numerical model, showing close agreement with microseismic event trends.

Field Trial Results. As of the writing of this paper, XP3's daily fluid production increased from 6.18 t to 23.07 t post-refracturing, and daily oil output rose from 1.50 t to 11.45 t. Water cut dropped from 75.7% to 46.3%. Post-treatment production levels returned to those seen during the initial well completion, and the recovery factor is expected to increase from 4.0% to 11.3%, indicating a 7.3% improvement in ultimate recovery. This demonstrates the effectiveness of the integrated casing reconstruction and refracturing approach in accessing residual reserves between fractures and adjacent wells.

Conclusions

This study addresses issues in the West 233 block of the Ordos Basin such as low liquid production, insufficient initial stimulation, wellbore integrity failure, and low formation energy. Taking well XP3 as the research object, targeted engineering logging, microseismic monitoring, pressure recovery, and fiber-optic injection profiling tests were conducted. These clarified the wellbore failure conditions, low degree of initial fracturing, low formation energy, and heterogeneous water coning in the wellbore, laying a foundation for numerical simulation and optimization of refracturing parameters.

Based on the test results, a casing-in-casing simulation was performed to optimize energy supplement parameters, fracturing cluster placement, and refracturing operation parameters. The optimization results indicate that well XP3 meets the conditions for full wellbore small casing reconstruction. The optimized

refracturing parameters can create symmetrical bilateral fractures with an average half-length of 158.6 m, increasing the stimulated reservoir volume (SRV) to 87.5%.

Field tests show that the refractured fractures have half-lengths between 150 and 200 m, averaging 158.6 m, generating a large-scale SRV signal and achieving the designed stimulation effect. After refracturing, daily oil production of well XP3 increased significantly from 1.5 t to over 10 t.

Acknowledgement

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References

1. Li Yang, Zhao Qingmin, Lv Qi, et al Development Evaluation Technology and Practice of China's Continental Shale Oil [J]. *Petroleum Exploration and Development*, 2022, **49**(5): 955–964.
2. He Yonghong, Xue Ting, Li Zhen, et al Technical Practice of Chang 7 Shale Oil Development in Ordos Basin—A Case Study of Qingcheng Oilfield [J]. *Petroleum Exploration and Development*, 2023, **50**(6): 1245–1258.
3. He Yonghong, Li Zhen, Fan Jianming, et al Well Pattern Optimization Technology and Practice for Shale Oil Development in the Ordos Basin—A Case Study of Qingcheng Oilfield [J]. *Acta Petrolei Sinica*, 2024, **45**(4): 683–697.
4. Zhang Kuangsheng, Mu Lijun, Lu Hongjun, et al Overview and Practical Understanding of Hydraulic Fracturing Test Site Construction for Shale Oil in the Ordos Basin [J]. *Drilling & Production Technology*, 2024, **47**(6): 16–27.
5. Guo Jianchun, Ma Li, Lu Cong. Advances and Development Directions of Fracture and Oil-Driving Technologies in Tight Oil Reservoirs in China [J]. *Acta Petrolei Sinica*, 2022, **43**(12): 1788–1797.
6. Shi Daohan, Zhang Kuangsheng, Tang Meirong, et al Development and Application of Volume Fracturing Technology for Shale Oil Horizontal Wells in Changqing Oilfield [J]. *Petroleum Science & Technology Forum*, 2022, **41**(3): 10–17.
7. Fang Maojun, Du Xulin, Bai Yuhu, Li Hao, Zhang Hao, Zhu Haiyan. Three-Dimensional Physical Simulation Experiment of Large-Scale Hydraulic Fracturing in Multi-Layer Tight Sandstone Reservoirs [J]. *Petroleum Experimental Geology*, 2024, **46**(4): 786–798. doi:[10.11781/sysydz202404786](https://doi.org/10.11781/sysydz202404786)
8. Yang Yonghua, Song Yanggao, Wang Xingwen, Liu Lin, Ci Jianfa, Lin Lishi. Hydraulic Fracturing Practices and Understanding of Weirong Shale Gas Field [J]. *Petroleum Experimental Geology*, 2023, **45**(6): 1143–1150. doi:[10.11781/sysydz2023061143](https://doi.org/10.11781/sysydz2023061143)
9. Bai Xiaohu, Qi Yin, He Shanbin, et al Integrated Refracturing Technology of Fracturing-Energy Supplementation-Oil Displacement for Tight Reservoir Horizontal Wells [J]. *Fault Block Oil & Gas Fields*, 2021, **28**(1): 63–67.
10. Xie Junhui, Wang Chengping, Yin Hui, et al Research and Application of Refracturing Technology for Old Wells in Jimusar Shale Oil Field [J]. *Petroleum & Natural Gas Chemical Industry*, 2021, **50**(4): 100–103.
11. Liu Yaowen, Ming Yue, Zhang Xudong, et al Mechanical Isolation Refracturing Technology of Casing-in-Casing in Fuling Shale Gas Wells [J]. *Petroleum Drilling Techniques*, 2022, **50**(3): 86–91.

12. Zhang Liwei, Li Jie, Yu Huiyong, et al Research and Field Tests of Horizontal Well Casing Expansion and Wellbore Reconstruction Technology [J]. *Drilling & Production Technology*, 2024, **47**(5): 67–73.
13. Fu Jinhua, Li Shixiang, Niu Xiaobing, et al Geological Characteristics and Exploration Practice of Triassic Chang 7 Shale Oil in Ordos Basin [J]. *Petroleum Exploration and Development*, 2020, **47**(5): 870–883.
14. Xi Kelai, Li Ke, Cao Yingchang, et al Organic-Rich Shale Lamina Assemblage and Shale Oil Enrichment Model of Triassic Yan'an Formation Chang 7 Subsection in Ordos Basin [J]. *Petroleum Exploration and Development*, 2020, **47**(6): 1244–1255.
15. Guo Xiujuan, Wang Jianning, Wu Yue, et al Controlling Factors and Distribution Characteristics of Oil-Bearing Tight Oil Reservoirs—A Case Study of Chang 8 Oil Layer Group in Red River Oilfield, Ordos Basin [J]. *Fault Block Oil & Gas Fields*, 2024, **31**(1): 69–76.
16. Fu Jinhua, Li Shixiang, Guo Qiheng, et al Enrichment Conditions and Favorable Area Selection of Continental Shale Oil in Ordos Basin [J]. *Acta Petrolei Sinica*, 2022, **43**(12): 1702–1716.
17. Li Shanshan, Sun Hu, Zhang Mian, et al Segmented Cluster Fracturing Technology for Shale Oil Horizontal Wells in Longdong Area of Changqing Oilfield [J]. *Petroleum Drilling Techniques*, 2021, **49**(4): 92–98.
18. Yang Shangyu, Yan Yan, Han Lihong, et al Research Progress of Full Lifecycle Wellbore Integrity Technology for Shale Oil and Gas Wells [J]. *Petroleum Pipe Materials & Instruments*, 2024, **10**(3): 1–6, 111–112.
19. Zhang Kuangsheng, Qi Yin, Xue Xiaoji, et al CO₂ Regional Energy Supplement Volume Fracturing Technology for Shale Oil Horizontal Wells in Ordos Basin [J]. *Petroleum Drilling Techniques*, 2023, **51**(5): 15–22.
20. Cai Meng, Tang Pengfei, Wei Xu, et al Optimization of Composite Volume Fracturing Technology for Gulong Shale Oil in Songliao Basin [J]. *Daqing Petroleum Geology & Development*, 2022, **41**(3): 156–164.
21. Joslin K, Ranjbar E, Shahamat S, et al Pressure Sink Mitigation: Effect of Preloading Parent Wells to Better Control Infill Hydraulic Fracture Propagation[C]. SPE Canada Unconventional Resources Conference. SPE, 2020: D023S001R002.
22. Shah M, Shah S, Sircar A. A Comprehensive Overview on Recent Developments in Refracturing Technique for Shale Gas Reservoirs [J]. *Journal of Natural Gas Science and Engineering*, 2017, **46**: 350–364.